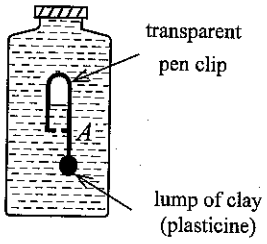
21. When a small amount of Benedict solution is added to a certain food extract and heated, no colour change occurred. When a little of dilute sulphuric acid is added to it and heated, the solution turned orange. The nutrient contained in this food extract would be,
(1) glucose. (2) sucrose. (3) fat. (4) protein.

22. Consider the following characteristics possessed by a living species.
A - An organized nucleus is absent.
B - can be destroyed by antibiotics
C - can be either pathogenic or autotrophic
Of these, the characteristics owned by the organisms belonging to domain bacteria are,
(1) only *A* and *B*. (2) only *A* and *C*. (3) only *B* and *C*. (4) all *A*, *B* and *C*.

23. The metals *P* and *Q* react with dilute acids liberating hydrogen gas while the metal *R* does not liberate hydrogen with dilute acids. *P* cannot displace *Q* from a solution of a salt of *Q*. What is the sequential order in which the metals *P*, *Q* and *R* are placed in the activity series?
(1) *P*, *Q*, *R* (2) *Q*, *P*, *R* (3) *P*, *R*, *Q* (4) *R*, *P*, *Q*

24. An arrangement named *A* has been introduced into a well stoppered plastic bottle filled with water so that it floats immersed. When the bottle is squeezed at its middle the arrangement *A* moves down the bottle. The reason for this is,
(1) increase in the pressure of water.
(2) increase in the weight of arrangement *A*.
(3) decrease in the upthrust exerted on *A*.
(4) decrease in the volume of water.

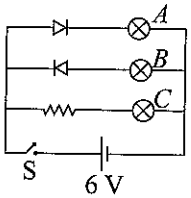


25. Given below are three statements relating to motion of waves.
A - Transverse waves travel in a direction perpendicular to the direction of the vibration of particles in the medium.
B - Longitudinal waves can travel in free space (vacuum).
C - Electromagnetic waves travel at the same speed in any medium.
The correct statement/s of the above is/are,
(1) only *A*. (2) only *B*. (3) only *A* and *B*. (4) only *B* and *C*.

26. Of the two solutions *A* and *B*, no colour change occurred when a few drops of phenolphthalein was added to solution *A*. When equal volumes of solutions of *A* and *B* of equal concentration were mixed and allowed to react, an aqueous solution of *C* was obtained and its pH was 7. The solutions *A*, *B* and *C* respectively could be,
(1) HCl, NaOH and NaCl. (2) HCl, NaCl and NaOH.
(3) NaOH, HCl and NaCl. (4) NaOH, NaCl and HCl.

27. The carbohydrate products first formed during photosynthesis, stored temporarily in chloroplasts and translocated through the phloem respectively are,
(1) monosaccharides, disaccharides and polysaccharides.
(2) monosaccharides, polysaccharides and disaccharides.
(3) disaccharides, polysaccharides and monosaccharides.
(4) polysaccharides, monosaccharides and disaccharides.

28. When the switch S is closed in this circuit,
(1) bulbs *A* and *B* light. (2) bulbs *B* and *C* light.
(3) bulbs *A* and *C* light. (4) all the bulbs light.



29. The number of autosomes and the sex chromosome that can be present in a human ovum respectively are,
(1) 22 and *Y*. (2) 22 and *X*. (3) 23 and *Y*. (4) 23 and *X*.

30. What is the option which correctly indicates how the curvature of the diaphragm and the volume of the thorasic cavity change when inspiration occurs in the respiratory process of a human?

	curvature of the diaphragm	volume of the thorasic cavity
(1)	increases	increases
(2)	increases	decreases
(3)	decreases	increases
(4)	decreases	decreases

[*See page four*